HCSA Rev D design spec (14 Oct 2026)

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The one source for the Rev D blueprints, the TRUTH Rev D revision and the website. Units mm, kN, MPa, kg unless said. Status tags: LOCKED (unchanged from TRUTH Rev S), FEA (set by the 9 to 12 Oct runs: OpenSeesPy v1, CalculiX, sfepy, independently reviewed), PROPOSAL (a Rev D detail drawn to carry an FEA load; sized in the Ansys submodels on Fri 16 Oct and in a physical test), OPEN (a decision Manos has not taken; Rev D states the default and the alternative).

Why Rev D exists: the 10 Oct whole-shell runs showed Rev S cannot hold 1 atm (31 mm glass at 54 MPa and 7.3 mm, ribs far over yield, open corners putting 100 to 138 kN on end bolts that break at 34 kN). TRUTH 9.5’s own decision tree says that if a thicker 6-pane L7 still fails stress, the 24-pane subdivision becomes the architecture. That branch is now taken. Everything else that passed in every whole-shell run (corner node, seam mid hardpoints, 375 mm main ribs, 25.4 mm glass, bolts as seal clamps) is adopted here as the Rev D baseline, so that the tests from November on test the design we mean to build.

1. Geometry (LOCKED, unchanged)

Item	Value
Topology	truncated icosahedron: 20 hexagons, 12 pentagons, 90 seams, 60 vertices (every vertex is hex-hex-pent)
Edge	2,250
Outer diameter / circumradius	11,151 / 5,575.5
Hexagon	4,500 corner to corner, 3,897 across flats, 13.153 m2
Pentagon	3,828 circumscribed diameter (1,914 circumradius), 1,548 inradius, 8.710 m2
Dihedral	hex-hex 138.19 deg, hex-pent 142.62 deg
Joint planes	radial (contain the seam and the shell centre); miter hex 20.905 deg, pent 16.472 deg
Internal pressure	101.325 kPa; hexagon face load 1,333 kN, pentagon 883 kN
Seam membrane tension	276.7 kN/m in the model (pR/2 bound 282.5); 623 kN per 2.25 m seam
Shell surface	367.574 m2

2. Hexagon panel: 24 panes (FEA)

- 6 main ribs, centroid to each vertex, 2,250 long, each split at its mid-point.
- 18 secondary ribs, all 1,125 long: 6 join the main-rib mid-points into an inner hexagon of 1,125 edge; 12 join each main-rib mid-point to the two neighbouring seam mid-points, so two secondary ribs of each panel land at every seam mid-point.
- 24 equilateral triangular panes of 1,125 side (full triangle, the span the FEA models; the glass itself is cut to about 1,062 side as in the shell model, and the clear opening between the 200 / 64 / 50 mm bezels is smaller: the transparency-led rib study sets it), inradius 325, 0.548 m2 each, 55.5 kN each at 1 atm. 480 panes on the shell (Rev S: 120 of 2.19 m2).
- Aluminium per hexagon: frame 13.5 m, main ribs 13.5 m, secondary ribs 20.25 m.
- Rev S (6 panes of 2.19 m2, 31 mm glass) is superseded: it fails stress (54 MPa vs 35) and deflection (7.3 mm vs 3.1).

3. Ribs (FEA; sections as I-beams, depth into the cabin from the outer face)

Member	Section	Stress at 1 atm	Note
Main rib, hexagon	I 375 deep, flanges 200 x 15, web 8; A 8,760 mm2; I 2.22e8 mm4	106 to 128 MPa	319 kg per hexagon; by hand (10 Oct verification) 275 mm is the shallowest that passes and 250 fails
Main rib, pentagon	I 400 deep, flanges 80 x 25, web 8; A 6,800 mm2	114 to 140 MPa	5 per pentagon, centroid to vertex, 1,914 long

Secondary rib	I 160 deep, flanges 64 x 8, web 6; A 1,888 mm2	116 to 147 MPa with the mid hardpoint pinned (Rev D detail, section 6)	if the hardpoint detail turns out moment-carrying the stress is 184 to 186 and the section becomes I 200, 100 x 10 (OPEN until C3)
Allowable	160 MPa (6061-T6 yield 240 / 1.5); local buckling b/t <= 10 flange outstand, h/t <= 50 web		all three pass both

Rev S ribs (I 103 deep, 64 x 10 flanges, 12 web) are superseded: about 1,470 MPa in the Rev S whole-shell run (v0, 9 Oct).

Factor-of-safety basis (OPEN, decision 10): the sizing runs used 160 MPa = 6061-T6 yield / 1.5 against the unfactored 1 atm stress, i.e. a proof factor of 1.5 on yield at MEOP. NASA-STD-5001 crewed pressure-structure practice is 1.5 proof and 2.0 ultimate on MEOP: on 2.0 against UTS 290 the allowable is 145 MPa, which the secondary rib (116 to 147) and the pentagon rib (114 to 140) sit at, and the main rib (106 to 128) clears. The next sizes (secondary I 200 / 100 x 10, pentagon rib deeper flanges) restore margin on that basis. Glass: the 35 MPa design allowable is TRUTH 9.5's conservative figure (a factor of about 4 on a ~140 MPa borosilicate strength); on NASA's factor of 3 against the measured characteristic strength, T0 has to return at least 81 MPa at the pane's effective area for the 27 MPa the pane sees. Agree the basis with Prof. Fleeter and Prof. Henann before the 8 Nov drawings.

4. Edge frame and seam (Rev S section kept; roles changed by FEA)

Section, per frame (A flat face, B grooved face; two frames per seam, 140 deep): - Web 20 x 140. Outer flange 50 x 10 at z 0 to 10. Seat lip 50 x 10 at z 93 to 103 (L7 seat). Depths z from the space face. - Frame B: outer O-ring groove centre z 18, inner z 32, gland 7.1 wide x 4.0 deep (Rev S 6.6; widened to Parker face-seal practice for a 5.33 cord: fill 78 %, squeeze 25 %) PROPOSAL; RTV566 recess R3 x 3 deep at z 42; leak port dia 2 at z 25 between the O-rings, one per 75 mm. - Bolt row z 58, 29 stations at 75 pitch (x 75 to 2,175; 30 collides at pentagon corners), hole 8.4. In Rev D stations 1 and 29 (x 75 and 2,175) fall inside the 100 mm corner blocks and are replaced by the vertex ties, and station 15 (x 1,125) is taken by the hardpoint pin: 26 seal-clamp bolts per seam, 2,340 on the shell (the FEA ran 29 per seam; the difference is inside the 5 % the bolts carry). PROPOSAL, confirmed in C1 and C3. - Frame stress 102 to 134 MPa (12 Oct FINAL run); frame bow between corners 1.4 to 2.0 mm against TRUTH's 1 mm sealing limit: OPEN (decision 5). Rev D keeps 140 deep and carries the bow limit as open until T1 and C2; a 300 mm frame still bows 1.07 (hand estimate). - No tongue-and-groove shear key in Rev D (Rev C CAD had a 14 x 24 tongue; the Rev S section the FEA and the T1 rig use has none). Shear along the seam goes through the vertex dowels and the hardpoint pin; the bolt row and the clamped faces carry the rest in friction. OPEN (decision 9): reinstate a key if the C2 and whole-shell shear checks need it.

Load path (FEA, whole shell with the corner node and hardpoints): the 623 kN per seam splits 63 % to the two corner nodes (196 kN per node end), 31 % to the mid hardpoint (152 to 195 kN across the two runs) and 5 % to the bolts (31 kN over the seam: about 1 kN of external tension on the average bolt, rising to 11 to 12 kN on the bolt next to each corner node; these are external loads from the seam, before the 10 kN preload). The bolts are seal clamps, not the tension path. On the Rev S path every bolt carries the whole seam tension: 20.7 kN each at the FEA's 276.7 kN/m (TRUTH Rev S had assumed 98.7 kN/m and 7.4 kN per bolt), and the slice then pries open at the seals at 0.3 atm and exceeds the 17 kN bolt limit at 0.46 to 0.73 atm (D1, prying and balanced pulls).

5. Corner node (FEA load; PROPOSAL detail)

- Each panel corner is one machined 6061-T651 block that includes the first 100 mm of both frames meeting at that corner (the FEA node: the first 100 mm of all six frames at a vertex, solid). 5.8 kg per vertex, 350 kg on the shell (three blocks per vertex, one per panel).
- Loads per panel corner at 1 atm: hexagon 380 to 430 kN with 50 to 65 kN m; pentagon 379 kN with 71 kN m. Node stress 130 MPa (limit 160). Stiffness 4e6 to 5e6 N/mm, 4e10 to 1e11 N mm/rad.
- Rev S corner as drawn (frames meeting on the miter, open) is 140 to 480 times too soft in translation and 4 to 9 times in rotation: superseded.
- Vertex tie (PROPOSAL, sized in C1 on 16 Oct): across each of the three joint faces at a vertex, 4 x 5/8-18 UNF A286 bolts through the two blocks in two rows (z 58 and z 115; the block is solid there, the pane corners are clipped well past it), for the 196 kN of direct seam tension per face (four bolts break at 591 kN on 165 mm2 each). The ties are sized for tension only. The 50 to 65 kN m per panel corner is not carried by the bolts: it goes through the block-to-block face contact across the joint plane (compression on one side, the ties on the other) and into the frames and ribs beyond, and C1 is the run that shows that path and sizes it. Dowels dia 12 x 2 per face carry shear along the seam and align the blocks. Corner seal: the three seal loops end at a vertex

plug on the radial line (Rev C item N1), still OPEN (O10).
- Main rib ends land on the block (rib-to-frame corner joint, failure mode 7).

6. Seam mid hardpoint (FEA load; PROPOSAL detail)

- At the mid-point of every seam (90 on the shell), where four secondary ribs land (two from each panel).
- Lug on frame A (40 thick) between a clevis on frame B (2 x 25), overlapping across the joint plane; pin dia 40 A286 along the seam (x) so the joint carries the 152 to 195 kN across the seam (10 and 12 Oct runs) and no moment about the seam. Hand checks at 195 kN: pin bending governs, M = P/2 x (t_clevis/2 + t_lug/4) = 2.19 kN m, 349 MPa on Z = 6,283 mm3 (a dia 30 pin would see 827 MPa, above A286 yield, so 30 is out); double shear 78 MPa; bearing 122 MPa in the lug, 98 in the clevis. Lug and clevis occupy z 50 to 135 so the three seal lines (z 18, 32, 42) run through uninterrupted above them. Station 15 has no clamp bolt: the block is clamped by the station 14 and 16 bolts at its ends (x 1,050 and 1,200). Sized in C3.
- Local block 150 long (x 1,050 to 1,200) in both frames, z 10 to 140; the four secondary rib ends bolt to it. About 8 kg per hardpoint over the frame section (PROPOSAL estimate), 720 kg on the shell.
- Stiffness at least 1e7 N/mm. Pinned is the Rev D intent; if the built detail is moment-stiff the secondary rib changes (section 3).

7. Glazing stack per pane (LOCKED layers, L7 by FEA)

Layer	Material	Thickness	z (from the space face)	Rev D change
L1 outer debris pane	fused silica	6	12 to 18, on 2 mm strips under the outer flange, cam locks	the main rib outer flange is 15 thick, so it is rebated 5 deep over the 32 mm seat band each side so the L1 strip sits at z 10 everywhere (about 5 % off the rib’s I, stress about 111 to 134 MPa, still under 160) PROPOSAL
L3 gap	dry N2 or Ar, IGU spacer	12	18 to 30	none
L4 PV	perovskite on 3 mm glass	3	30 to 33	none (concept stage)
L5 electrochromic	WO3 / NiO on 3 mm ITO glass	3	33 to 36	none (concept stage)
L6 shutter	7075-T7352 louvers, two 1.5 mm levels	3 envelope	44 to 47	none
L7 pressure pane	BOROFLOAT 33 borosilicate, fine annealed	25.4 (Rev S 31)	105 to 130.4 on a 2 mm strip on the seat lip, 4 mm retaining bead	FEA: 22.4 to 22.9 MPa on a flat hinge (27 mesh-corrected) against 35 allowable; bending 1.03 mm; 1.6 to 1.8 mm with seat movement against the 1.62 limit (1/200 of the 325 inradius)
L2 / L8 frames and ribs	6061-T651 (prototype), 2219-T851 (flight)	140 frame, 375 ribs		ribs now 375 deep into the cabin

- L7 25.4 is the thickest standard BOROFLOAT 33; the next step up is SUPREMAX 34.9 (SCHOTT, 9 Oct), which weighs 37 % more. Rev D keeps 25.4 with a stiffer setting strip (1.0 mm, 80 Shore A) to bring the seat movement inside the limit: OPEN (decision 2), settled by C4 and a full-size pane-on-seat test (T2 cannot settle it: its strips do not scale).
- Deflection limit basis: 1/200 of the inradius is 1.62 mm on the 1,125 triangle the model spans and 1.53 mm on the 1,062 cut pane. The FEA bending alone (0.8 to 1.03 mm) passes either; with seat movement (1.6 to 1.8) it misses either, which is why the seat is decision 2.
- Glass strength requirement (Weibull step D4): a pane failure probability of 1e-4 at 1 atm needs a characteristic disc strength of 412 / 176 / 94 / 57 MPa at m 5 / 7 / 10 / 15. T0 on bars measures m; the bar effective area is different and D4 is rerun for bars on 16 Oct.
- Pane edge: 10 mm clear of the web, 20 mm bite on the seat.

8. Pentagon panel (FEA loads and plate; rib layout PROPOSAL)

- 5 main ribs (section 3) centroid to vertex; 15 secondary ribs (5 forming the inner pentagon of 1,125 edge, 10 from rib mid-points to the two neighbouring seam mid-points, 957 long); 20 triangular bays. The shell model

carried the plate on secondary ribs; this exact layout mirrors the hexagon and is a PROPOSAL.
- Face plate 10 mm 6061-T6 (Rev S 4 mm), about 120 MPa mesh-corrected; 235 kg.
- Pentagons stay opaque: solar, shields, docking and services as in TRUTH 8.4, 11, 12. A 0.2 % pentagon oversize is a 140 MPa self-stress (12 Oct DS run; it is also the hand value 0.002 x 69 GPa): pentagon frames need the same machining tolerance as the hexagons (0.02 % evens the corner loads by 9 %).

9. Fasteners and seals

Item	Rev D	Status
Seam bolt	5/16-24 UNJF, A286 per AMS 5737 (130 ksi min, about 34 kN break), hex head, nut and two hardened washers; 10 kN preload, set by torque for 20 to 25 % squeeze from the 550 article; role: seal clamp, external seam load about 1 kN average and up to 12 kN next to the corner nodes; 26 per seam at the 75 pitch stations not taken by the corner blocks and the hardpoint (2,340 on the shell)	FEA role; designation OPEN: “NAS9306C-06” is a blind rivet standard and a 3/16 size, corrected in the 1 Nov parts memo (candidate NAS6605-type hex bolt)
Bolt length	grip 43.2 (two 20 webs, two 1.6 washers), 2-1/4 in (57) minimum	fixed 14 Oct (1-3/4 was short)
Vertex tie	4 x 5/8-18 A286 per joint face, two rows	PROPOSAL
Hardpoint pin	dia 30 A286, double shear	PROPOSAL
Primary seals	two 5.33 cords in 7.1 x 4.0 glands, 25 % squeeze: Parker BA500 butyl closed loop (Parker, 5 Oct: fluorosilicone permeates in vacuum, butyl does not, and butyl is the cord Parker can extrude for the -40 C service case) with FVMQ AMS-R-25988 as the alternative	OPEN (decision 3, before the 8 Nov drawings); TRUTH Rev S baseline was FVMQ
Tertiary seal	RTV566 bead R3 x 3, SS4155 primer, 10 mm inboard of the inner gasket	LOCKED
Thermal	-40 to +120 C seam sweep is benign: bolt 8.0 to 13.3 kN, squeeze 24.2 to 26.2 %; the cold limit is the elastomer’s (FVMQ -73 C, butyl -59 C)	FEA (D6)

10. Materials and heat treatment

Part	Material	Heat treatment	Properties used
Frames, ribs, corner blocks, hardpoint blocks, pentagon plates (prototype)	6061-T651 plate / 6061-T6511 extrusion	solution 530 C, water quench, stretch 1.5 to 3 %, age 160 C 18 h; never welded (HAZ loses 40 to 50 % of yield), never above 150 C	yield 240, UTS 290, E 68.9 GPa, CTE 23.6 ppm/K, allowable 160
Same parts (flight)	2219-T851	solution 535 C, quench, stretch, age 175 C 18 h	yield 352, UTS 455
Seam bolts, vertex ties, hardpoint pins	A286	solution 980 C, age 720 C 16 h	UTS 895 min (130 ksi), yield 655, CTE 16.5
Shutter	7075-T7352	solution 480 C, quench, stress-relieved, two-step age (about 107 C then 163 to 177 C)	yield about 435
L7	BOROFLOAT 33	fine annealed by SCHOTT; not tempered, not chemically strengthened	E 64 GPa, CTE 3.25, allowable 35 (design), strength from T0
L1	fused silica	annealed	E 72 GPa, CTE 0.5
Seals	butyl BA500 or FVMQ	post-cured by the maker	see section 9
Test rigs	4140 Q&T 28 to 32 HRC (blocks, adapters), A36 (tub), A2 58 to 60 HRC (rollers)	as in the rig pack Rev B	

11. Mass (FEA sections, first order; proposals for nodes and hardpoints included)

Item	Per panel	Shell
Hexagon: frame 124, main ribs 319,	1,838	36,760 (20)

secondary ribs 103, L7 745 (25.4 mm over the full 13.15 m2; the cut panes weigh about 664, so this is conservative by about 80 kg), L1 174, L4 88, L5 88, L6 111, hardware 50, node share 12, hardpoint share 24		
Pentagon: frame 103, main ribs 176, secondary ribs 77 (15 ribs, 15.2 m), plate 235, hardware 30, node share 10, hardpoint share 20 (base, before variant equipment)	651	7,810 (12)
Corner nodes 350 and hardpoint blocks 720 (both inside the panel figures as the shares above)		1,070
Structure only (frames, ribs, L7, plates, nodes, hardpoint blocks)		34.0 t (32.9 without the node and hardpoint proposals; Rev S 25.4 t, which fails; 34.9 mm glass adds 5.6 t)
Facility: hexagons 36,760 + pentagon bases 7,810 + pentagon variant equipment 3,300 (TRUTH 8.4: 6 x 190, 2 x 230, 710, 3 x 330) + stand-alone 2,480 (TRUTH 13.2)		50,350, about 50.4 t (Rev S 41.7 t)

Launch: Starship V3 (100 t conservative) keeps about 49.6 t margin; Falcon Heavy reusable (50 t) is over by about 0.4 t and is no longer a single-launch option; Falcon Heavy expendable (63.8 t) keeps about 13.4 t. The 25 t structure target of Rev S is withdrawn (decision 8). Mass comes down next from a transparency-led rib study (narrower, deeper ribs with a thicker web recover clear area at similar stress; curved glass is the untested hypothesis that could remove most ribs).

12. What success looks like, and what scales

The rule for every test: the digital twin files its prediction for every gauge before the test; the test passes when the measurement lands within 10 % at the first hold point; a miss changes the model, never the data; once a twin is trusted at test scale it is run at full scale and that is the scaling.

Test	Article	Scale and what scales	What does not scale, and the guard	Success
T0 glass strength (ENGN 1440, spring 2027)	BOROFLOAT 33 bars 1 in x 1/8 in, four-point bend, 30 per edge finish; alumina reference	Strength does not scale geometrically: it is statistical (Weibull). Pane strength = bar strength x (bar effective size / pane effective size)^(1/m), by effective area when the fractures start on the surface and by effective edge length when they start on the edge (four-point bars with ground edges usually do); both are recorded and the origin of every fracture is found	m is measured, not assumed; a second bar size checks the exponent between two sizes before it is trusted at 0.55 m2; edge finish classes separately	m and characteristic strength per finish; the predicted 1e-4 pane strength above the 27 MPa the pane sees (35 design); fractures start on the surface or edge as the finish class predicts
T1 seam slices (Instron 5882, after 18 Nov)	75 / 150 / 225 mm of the full-size seam, Rev D section	Full size; only the length is cut. Per-bolt response must be flat from 75 to 150 to 225 (the middle bolt of the 225 sees both neighbours, as every bolt in a seam does)	end dams, cut ends, grip legs: held by the D1 twin that models the slice and the 2,250 seam with the same code	At the Rev D service hold point, an applied pull of 12 kN per bolt (the worst bolt’s external load; 24 kN applied to the 150 slice on top of the 10 kN preload): inner gap under 0.266 mm (D1 predicts 154 um) and no leak; then up the D1 curve within 10 % to 0.55 atm of the Rev S upper bound; bolt-station stiffness measured (D1 70 to 80 kN/mm) and fed to the shell model
T2 one panel at 1:4 (water rig, spring 2027)	hexagon 1,125 corner to corner, 24 aluminium panes 6.0 mm, ribs 94 and 40 deep, 24 bays	Geometric similitude: at the same pressure a model with every length / 4 (thicknesses too) has the	setting strips and beads (rubber), bolt preload and friction, 2 mm webs (local buckling, machining),	every gauge within 10 % at 1 atm at the matching edge condition; linear and repeatable over three

		same stress in the ribs and frames and 1/4 the deflection; the aluminium panes match the glass plate stiffness only (D = E t3 / 12(1 - nu2): 6.35 mm glass, E 64, nu 0.2, equals 6.0 mm aluminium, E 69, nu 0.33), so their stress is not the glass stress; edges pinned and clamped bracket the real seat	fillets, gravity, and glass strength (never scales: T0 covers it). Guard: the panel model is built at both scales from one script; the items that do not scale are varied in the model to show they do not move the answer	cycles; the measured edge condition sits between pinned and clamped and is adopted in the full-scale model
T3 corner (2027)	one vertex: three corner blocks, six frame stubs 400 long, rib stubs, vertex ties	Full size. Load applied along the seats (never by gripping stub ends)	the seam shear and the dihedral moment need a three-actuator rig or a reacting fixture	rosettes at the junction within 10 % of C1; node stiffness measured and fed to the shell model
T1-550 article	550 mm straight seam, 7 bolts	Full size	not pulled	torque for 20 to 25 % squeeze; 1 atm between the seals holds for 1 h with the agreed decay
1:10 shell (only if the cap model disagrees with T2)	PLA or aluminium shell, 1,115 across	load sharing between panels, seams and corners is set by the geometry and is reproduced; stresses are not represented because the material and the thicknesses are not scaled	material	load distribution matches the shell model

Full facility against the tests: the facility is 11,151 across with 2,250 seams and 4,500 hexagons; T2 is the same hexagon at 1,125; T1 and T3 are the real seam and the real corner at 1:1; T0 is the real glass, scaled by statistics, not geometry.

13. Decisions still open (defaults in Rev D)

1. Hardpoint pinned (default) or moment-carrying: sets the secondary rib (160 or 200 deep). C3 on 16 Oct.
2. L7 25.4 with a stiffer seat (default) or 28 to 34.9: after C4, a full-size pane-on-seat test, and T0.
3. Seal cord: butyl BA500 closed loop (default) or FVMQ. Before 8 Nov.
4. Seam bolt designation. 1 Nov.
5. Frame bow limit 1 mm (TRUTH) against 1.4 to 2.0 predicted: keep 140 deep and test (default), or deepen. After T1 and C2.
6. Corner seal at the vertex plug (O10).
7. Vertex tie (tension only; the corner moment path through block contact) and hardpoint pin dia 40: C1 and C3 on 16 Oct.
8. Mass: Rev D structure 34.0 t, facility about 50.4 t; the transparency-led rib study is the next pass.
9. Shear key: none in Rev D (dowels and hardpoint pin carry seam shear); reinstate if C2 or the whole-shell shear check needs it.
10. Factor-of-safety basis: 1.5 on yield (used) or NASA-STD-5001 1.5 proof / 2.0 ultimate on MEOP; glass factor 3 on measured strength. With Fleeter and Henann before 8 Nov.